

CreditTrax

*Empowering Young Adults for Financial
Success*

Prototype Demo 1 Presentation
By Team: Copper
Old Dominion University
CS 411W
Spring 2025



Table of Contents

Title Page	1	User Stories to Features	21-26
Team Biography	3	User Risks	27
Problem Statement	4	Technical Risks	28
Our Solution	7	Issues and Challenges	29
RWP MFCD	10		
Prototype MFCD	11		
Database Schema	12		
Development Tools	13		
Required Libraries, Tools, & Tech	14		
Github Repository	15		
Sprint Breakdown	16-19		
Changes to Scope	20		

Meet the Team



Darrick Allen
Webmaster /
Frontend Developer



Brian Nguyen
Project Lead



Bryce Johnson
Backend Developer



Indi Monson
Backend Developer



Erika Andrade
Backend Developer



Josh Martin
Database Specialist



Smit Dalia
Documentation Specialist

Our Problem Statement

Many young adults in the U.S. struggle with their finances due to the overwhelming burden of student loans, ever increasing credit rates, and a lack of experience with crucial financial skills.

Balancing the pressures of starting a career, taking care of themselves and loved ones, all while facing financial uncertainty leads to the accumulation of debt, increased anxiety, and delayed life milestones such as marriage, home ownership, and starting a family.



The Problem Characteristics

Forgetting to pay monthly payments due to being busy

- ❖ Interest rate increase
- ❖ Takes even longer to pay it off
- ❖ Late fees
- ❖ Credit score sinks
 - Harder time making large purchases
- ❖ Mental and physical health decline
- ❖ Harder time achieving life goals.

Lack of proper budgeting skills and/or financial literacy

- ❖ Can't make realistic budgets
- ❖ Won't know where their money has gone
- ❖ Harder time prioritizing expenses, short-term and long-term goals
- ❖ Won't fully understand the impacts of late payments and general financial choices

Some apps have budget management features, some have debt management, not many have both

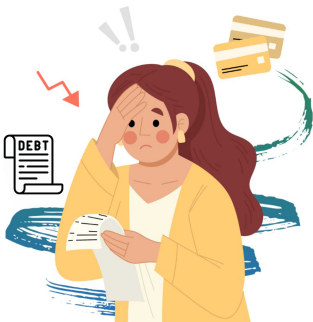
- ❖ May have to use at least two financial apps.
- ❖ Harder to manage finances
- ❖ Hard to see the bigger picture



Who is Affected?

Users

- ❖ **Young Professionals** overburdened by student loans and credit card debt.
- ❖ **Students and Recent Graduates** lacking financial literacy and seeking budgeting support.
- ❖ **Young Adults** (18-34) mentally overwhelmed by a lack of experience with finance.
- ❖ **Service members** transitioning from military to civilian struggling managing finances



Customers

- ❖ **Young Adults**
- ❖ **Financial Institutions** looking to partner with apps to better serve a younger audience.
- ❖ **Employers** looking to offer financial wellness programs for early-career employees.
- ❖ **Universities/Finance Professors** that want to offer an educational finance tool for students to learn personal finance



Stakeholders

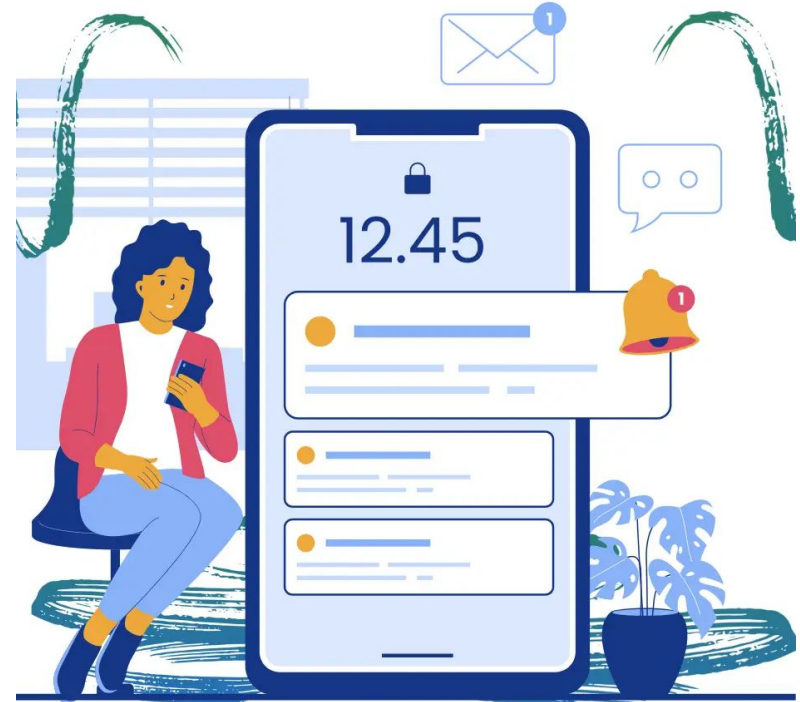
- ❖ **Financial Advisors** with an interest in helping young adults build sound financial habits
- ❖ **Financial Wellness Coaches** providing educational services to young adults to reduce debt and improve budgeting skills.
- ❖ **Credit Counselors** looking for tools to suggest to their younger clients.



Our Solution: CreditTrax

CreditTrax leverages AI-powered budgeting through personalized financial dashboards and interactive tools to help users align their spending with long-term financial goals while making smarter financial decisions.

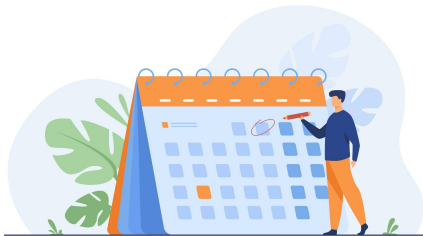
By analyzing spending habits, sending real-time notifications, and offering comprehensive financial literacy tools, CreditTrax enables young adults to achieve their financial objectives, track their progress, and reduce the stress and anxiety that often comes with managing debt.



Solution Characteristics

Debt Tracking

- ❖ Users can **input all their debts** e.g. credit card balances and student loans along with their respective interest rates and due dates
- ❖ **Payment Reminders** prompts the user to **manually approve payment** of a debt.



Budget Recommendations

- ❖ Powered by **generative chatbot AI** to provide personalized recommendations based on a user's debt and spending habits
- ❖ Analyzes the user's income, expenses, and debt obligations to generate tailored suggestions to help them optimize their budget

Personal Financial Dashboard

- Data analytics** - allows the dashboard to offer real-time visualizations and reports on financial health.
- Visualization** - uses charts/graphs and badges to illustrate the user's progress with their debt.

Encourages users to stay motivated to adhere to their budget and debt repayment plans.

- ❖ **"What-if" analysis** - This feature allows users to simulate various financial scenarios, such as modified loan payments or savings adjustments, to project outcomes and make informed decisions.

e.g. For users who manage to make every payment before the due date for 6 months straight.



Solution Characteristics

Financial Literacy

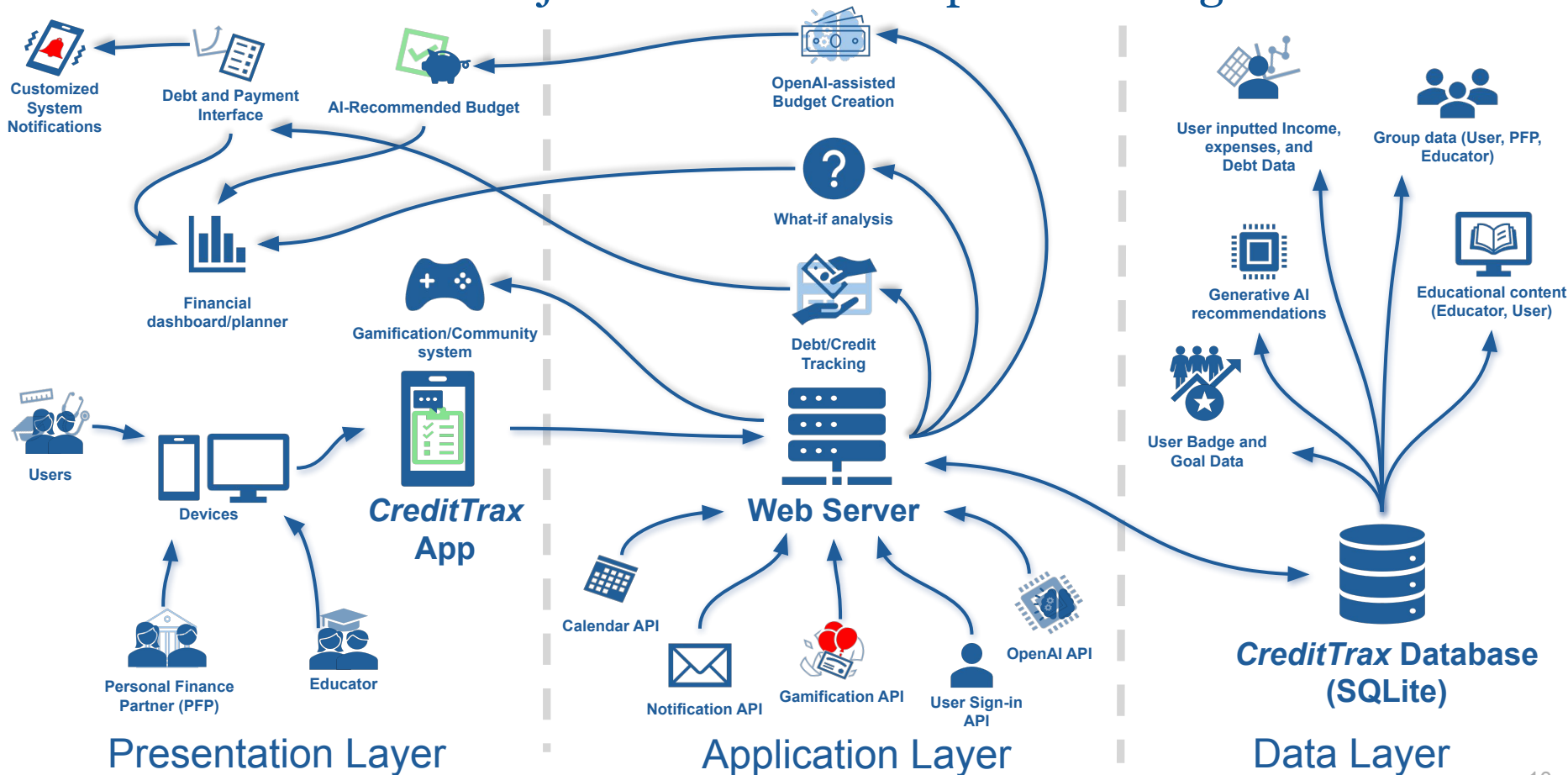
- ❖ **Builds essential financial skills:** CreditTrax educates users on key financial concepts like budgeting, saving, and managing debt, which are crucial for young adults who may lack experience in these areas.
- ❖ **Encourages proactive financial habits:** The app fosters good financial habits by teaching users how to balance debt payments with saving goals, enabling them to take control of their finances early in their careers.
- ❖ **Reduces financial anxiety:** CreditTrax helps young adults reduce financial stress by providing essential financial literacy skills, enabling them to effectively manage student loans and credit card debt.

Short vs Long-term Goals

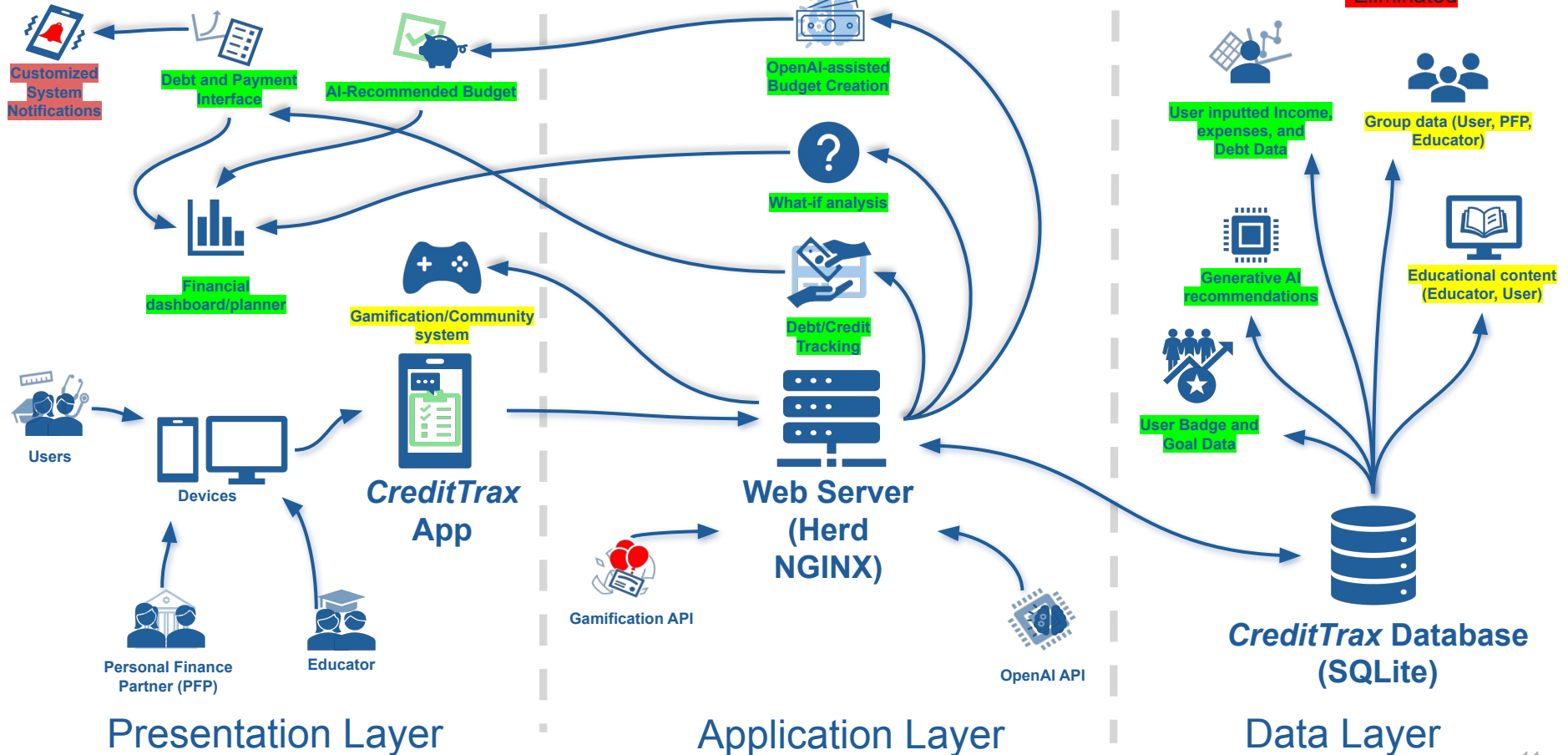
- ❖ This app offers the user the flexibility of choosing between setting short-term or long-term goals to suit the user's changing needs.
- ❖ They can choose from a list of options or define their own goals, e.g. short term: building an emergency fund or eliminating debts and long term: buying a house or a car.



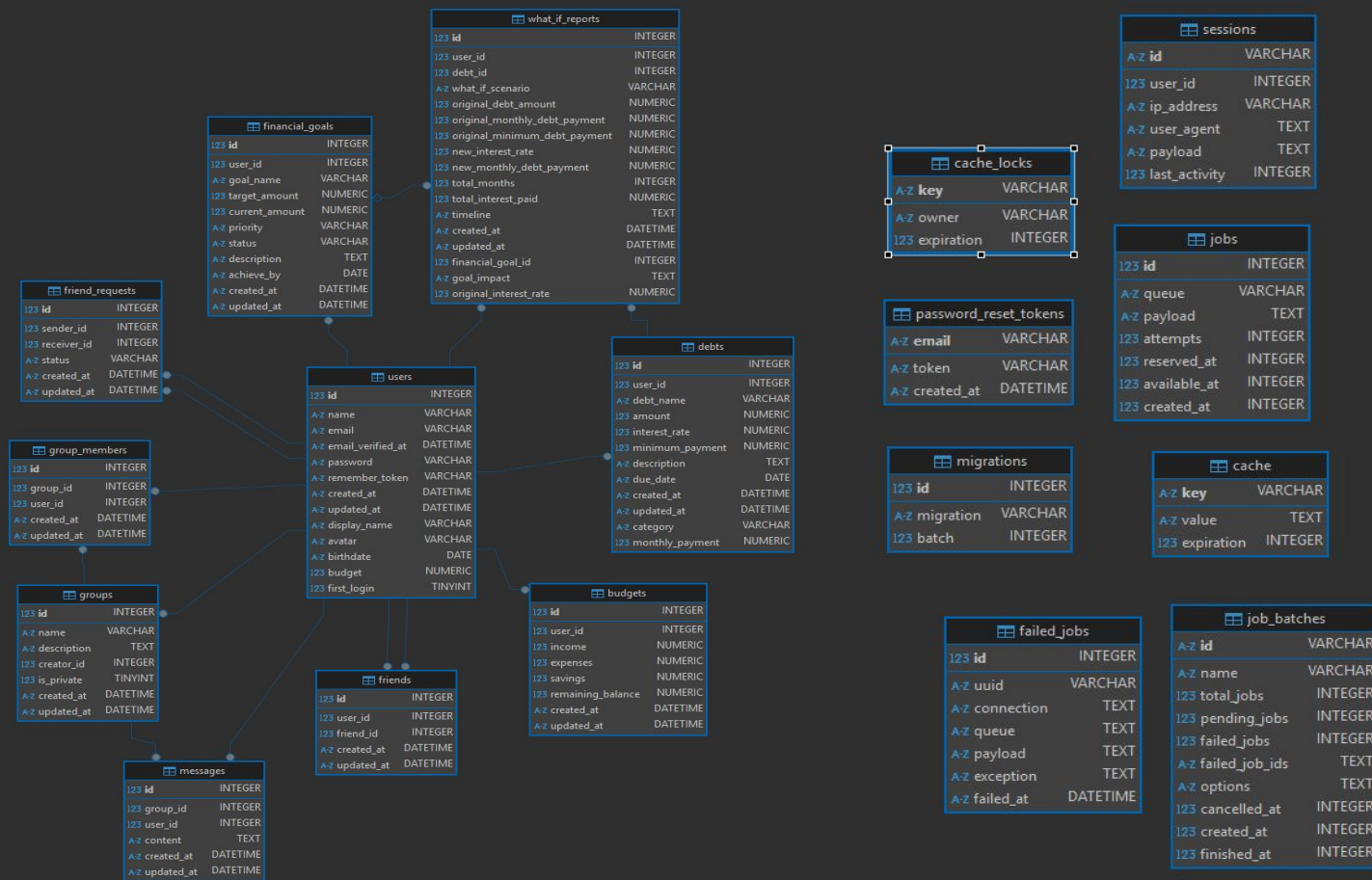
RWP Major Functional Components Diagram



Prototype MFCD Diagram



Database Schema



Development Tools

Development Type	Front-End	Back-end
IDE	VSCode	VSCode
Version Control	Github	Github
CI and CD	Github	Github
Language	HTML5, CSS, JS	PHP
Testing Framework	Jest	PHPUnit
Documentation Tool	JSDoc	Scribe

Required Libraries, Tools, & Technologies



Libraries

- Chart.js



Tools

- Herd: Local Development Environment Manager
- Laravel: Backend Framework for PHP
- TailWind: Frontend Framework for CSS
- Alpine: Frontend Framework for JS
- Composer: Dependency manager
- SQLite: Database Management System
- Laravel Forge: Deployment Service



Technologies (Hardware and Software)

- OpenAI API
- GameLayer API

Github Repository, Team Website, and Prototype URL

Everyone in Team Copper, Mr. Jacob Cook, and Professor Hosni should have access to the repository.

Github Repository: <https://github.com/bnguyen117/s25-Team-Copper>

Team Website: <https://dalleodu.github.io/Team-Website-2024/>

Prototype URL: <https://prod.s25-team-copper.stspreview.com/>

Sprint Breakdown

PROCESS	SPRINT 0		SPRINT 1		SPRINT 2		SPRINT 3		SPRINT 4		SPRINT 5		SPRINT 6	
	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Initial setup														
Set up Framework														
User Interface														
Feature Development														
System and Unit Testing														
Polish and Deployment														

Demo 2

Sprint Breakdown cont.

❖ Sprint 0 (Week 1-2)

- Setup development environment
- Create Trello board
- Create git repository

❖ Sprint 1 (Week 3-4)

- Set up frontend and backend frameworks
- Set up testing tools
- Setup APIs (Mostly OpenAI)
- Write initial unit tests
- Begin creating user stories

❖ Sprint 2 (Week 5-6)

- Develop basic UI functionality
 - Button presses
 - Page movement
- Begin developing core features:
 - What-if analysis, budgeting, debt tool
- Expand unit test

Sprint Breakdown cont.

❖ Sprint 3 (Week 7-8)

- Continue development of core features
- Begin development of other features
 - Dashboard, game, other users
- Finish creating unit tests
- Create/conduct performance tests
- Demo 1 in Week 7

❖ Sprint 4 (Week 9-10)

- Implement full functionality of features
- Create/conduct system tests

❖ Sprint 5 (Week 11-12)

- Finalize features
- Finalize UI functionality and form
- Review testing data
- Demo 2 in Week 11
- Changes:
 - Finish implementation of budget, debt tables, and rewards

Sprint Breakdown cont.

- ❖ Sprint 6 (Week 13-14)
 - Create placeholder content for demo
 - Beta-test
 - Publish and present
 - Prototype Demo

Changes to scope, implementation language, etc.

Demo 1

- ❖ Shifted the focus from AI-assisted financial/debt management app to an AI-powered **educational tool** to teach financial literacy and give recommendations.
 - Still will implement the AI-assisted budgeting. Primary focus will be in what-if analysis
- ❖ Changed development tools and backend programming language
 - **Python → Laravel PHP**
 - **Added frameworks for HTML, CSS, JavaScript compatible with Laravel**

Demo 2

- ❖ Eliminated "Smart Payment Reminders" feature for debt tracking
- ❖ Eliminated implementation Educator and Personal Finance Partner roles and their features.
- ❖ Our mentor, Jacob Cook, created a Laravel Forge server to deploy the prototype.

User Stories to Features

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Debt Tracking	Add, Edit, Delete and Track Debt	Create and edit a debt source with source name, amount due, interest rate, term length, and more	Young adults
		Delete a debt source	Young adults
		Create and edit a new debt transaction	Young adults
		Delete a debt transaction	Young adults
		See all my debt information condense into one easily accessible place	Young adults
	Confirm payment of debt in app	Confirm payment of a debt once I paid the actual source in real life.	Young adults
AI-assisted Budgeting	Standard budget creation	Create a budget without the use of AI if I don't want to use it	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
AI-assisted Budgeting	Create/edit/delete expense and category	Create, edit, and delete an expense transaction	Young adults
		Create, edit and delete an expense category	Young adults
		Create and edit a new income transaction	Young adults
		Delete an income transaction	Young adults
		Create, edit, and delete an income category	Young adults
	Track expenses	See the total amount of money I've spent on a category	Young adults
	AI-driven budget recommendations	Receive AI-driven budgeting suggestions, so that I can optimize my spending and savings based on my financial situation	Young adults
	View the budget	View the budget that I created.	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Goal Tracking	Goal Progress Visualization	Track my short-term and long-term goals in one place	Young adults
		Automatically update my goals based on transactions logged into the app	Young adults
		Press a button confirming I have completed a goal	Young adults
		View all my completed goals	Young adults
	Short-Term/ Long-term Goals	Have short-term goals so I can keep track of my personal finances in small milestones	Young adults
		Have long-term goals so I can set big financial milestones	Young adults
What-If Analysis	AI-assisted simulation/ recommendations	Use OpenAI model to ask for specific scenarios that the app won't be able to answer	Young adults
		Use OpenAI model to provide recommendations based on results of the algorithmic scenario.	Young adults

User Stories to Features cont.

*Done

*WIP

*Planned

Functionality Group	Feature	User Story	User Role
What-If Analysis	Scenario Predictions and Visualizations	See monthly savings after all expenses.	Young adults
		See a graph that shows future debt payments over a period of time.	Young adults
	Algorithmic What-If Analysis	Be able to perform basic predictions on savings and debt without the use of OpenAI model	Young adults
Financial Dashboard	Comprehensive summary of financial standing	See my debt to income ratio to help me not overextend myself	Young adults
		See my current total spending for this month	Young adults
		See current trends, pattern, and statistics generated by OpenAI model from analyzing my data.	Young adults
	Overview of goals	See a short-term goal, long-term goal, and progress for each of them	Young adults
	View achievements	See my last completed achievement	Young adults
		See an achievement that is the closest to completion	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Financial Dashboard	Access other pages from Dashboard	Go to the My Finances Page from the dashboard	Young adults
		Go to the Rewards page from the dashboard	Young adults
		Go to my Debt Table from the dashboard	Young adults
		Go to my goals section from the dashboard	Young adults
Gamification	Achievements	Earn achievements by completing certain requirements/milestones	Young adults
		Have the badges level up one tier when I earn it	Young adults

User Stories to Features cont.

*Done
*WIP
*Planned

Functionality Group	Feature	User Story	User Role
Community/Group	Groups	Accept/Send friend requests	Young adults
		Join a group with my friends	Young adults
		Create a group	Young adults
	Resources/Information	See accurate financial concepts throughout the app	Young Adults

User Risks

User Risks

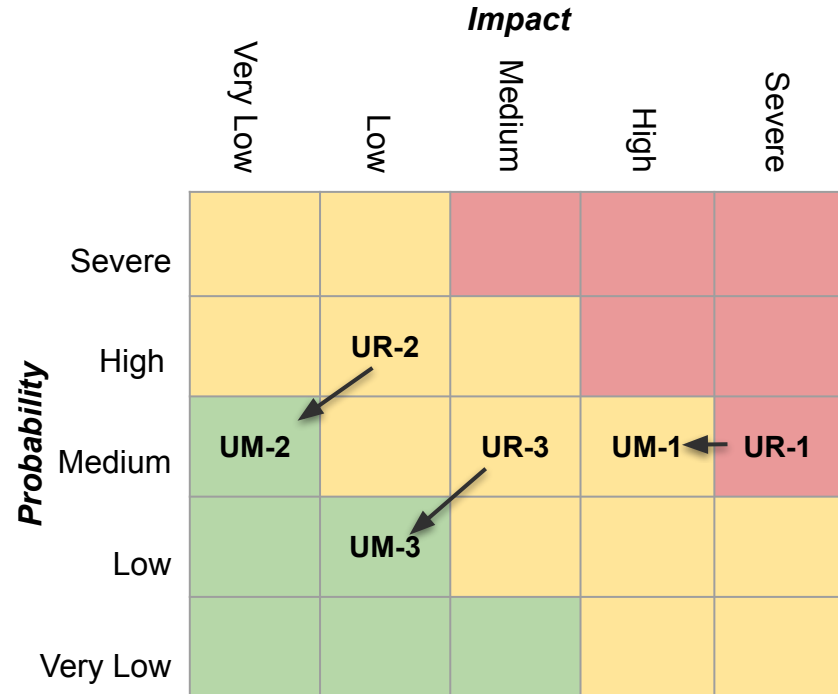
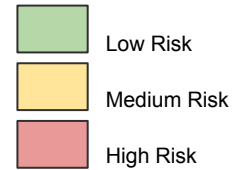
- **UR-1:** The tool's UI is too complicated, making data entry, tracking, and real-time management difficult and prone to errors.
- **UR-2:** User forgot to respond to CreditTrax when the monthly bill was paid.
- **UR-3:** User finds their current budget style unsatisfactory.

*Done
*WIP
*Planned

Risk Mitigation

- **UM-1:** Simplify the UI to enhance user experience, making data entry and tracking more intuitive and reducing the likelihood of errors. Provide tutorials or guided walkthroughs to help users navigate the app effectively.
- **UM-2:** Asks users if they paid the bill the next time they open the app.
- **UM-3:** CreditTrax can advise user another budgeting technique (50/30/20, zero-based, etc.)

Legend



Technical Risks

Technical Risks

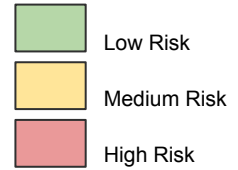
- **TR-1:** Integration of AI-driven budgeting recommendations and multiple features that can be complex
- **TR-2:** Ensure the app runs smoothly across different smartphones, operating systems, and versions.
- **TR-3:** Issues with handling concurrent users or processes at one time that also may make the application struggle to scale effectively.

*Done

*WIP

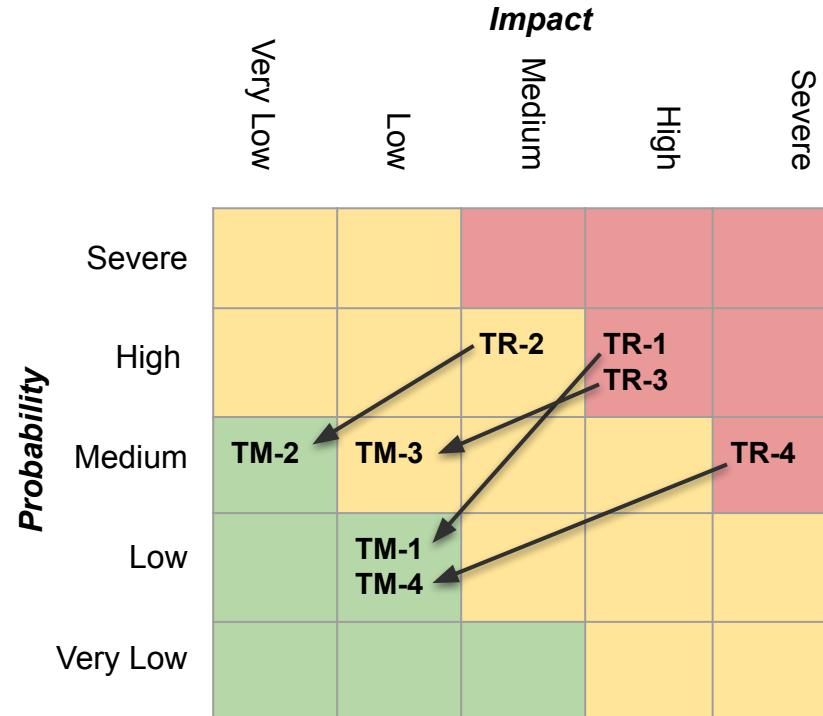
*Planned

Legend



Risk Mitigation

- **TM-1:** Break down the development process into smaller, manageable tasks; using modular architecture and thorough testing.
- **TM-2:** Develop a web application, testing fitment of UI across multiple screen sizes.
- **TM-3:** Implement concurrency control mechanisms and perform load testing with the use of scalable infrastructure and cloud services



Issues and Challenges

- Based on current understanding of OpenAI gpt-4o-mini, we cannot have AI create a budget based on user input.
- Experienced an CSRF mismatch error when attempting a POST request to GameLayer API
 - Switched to Laravel-Gamify, but encountering issues with having same column names in multiple migration files.
 - [qcod/laravel-gamify: Gamify your Laravel app with Reputation Points & Achievements Badges support](#)



Q & A

We welcome your
questions!

Next Steps: Brian

- Responsible for the Savings What-If Analysis Type
- Next Steps:
 - Finish the Savings Analysis Type, change database to reflect it.
 - Refactor the what-if analysis code to improve efficiency and readability.
 - Assist in creating the budget and improving upon it.
 - Assist in developing the rewards feature using laravel-gamify with Erika
 - Develop CRUD of debt transactions for a debt source.

Next Steps: Bryce

- Finish up touches on the dashboard
 - Make everything neat
- Finish the yes and no buttons on the dashboard with the debt graph.

Next Steps: Darrick

- Continue polishing What-If Analysis reports.
- Implement persistent chat-history for AI within a report
- Help implement the AI component for general budgeting suggestions not connected to a report.

Next Steps: Erika

- Refine Rewards / Achievements page by:
 - **Integrating laravel-gamify properly** so that it works as our badge system
 - Creating working badges for simpler events like visiting the debt page for the first time, etc.
 - Adding the functionality for badges to display their info when clicked

Next Steps: Indi

- Finish the community page
- So far the friends can request and accept and delete each other
- Next is the groups, with updating groups so they can display points from the rewards
- And messages each other

Next Steps: Josh

- Add a hard-coded budgeting algorithm based off of the 50/30/20 model of budgeting.
- Add a viewable budgeting breakdown to Finance page.

Next Steps: Smit

- Retrieve and Display Budget Data:
 - Fetch and display user-specific budget information from the database on the Budget UI.
 - Set up forms and backend logic to allow users to update and save their budget data.
 - Make the budget page responsive and improve user interaction with visual feedback and cues.

Appendix

...

User Stories

As a *Young Adult*, I would like to...

Financial Dashboard:

- Track my financial goals using a personalized dashboard so that I can stay motivated and reduce financial stress.

What-if analysis:

- Run a "what-if" analysis on my debt repayment strategies so that I can understand the long-term impact of different financial decisions.
- Use an algorithmic what-if analysis for simple financial scenarios.
- Use AI to analyze and generate recommendations based on results of an algorithmic what-if analysis.

Debt tracking

- Create a debt source with source name, amount due, interest rate, term length, and more
- Edit a debt source with source name, amount due, interest rate, term length, and more
- Delete a debt source
- Receive reminders for my payment due dates, so that I never miss a payment and avoid late fees.

User Stories

As a *Young Adult*, I would like to...

Debt-tracking cont:

- Add, edit, and delete new debt transaction
- See all my debt information condensed into one easily-accessible place.
- Confirm payment of a debt in the app once I paid the actual source in real life.

AI-assisted Budgeting:

- Receive AI-driven budgeting suggestions, so that I can optimize my spending and savings based on my financial situation.
- Create a budget without the use of AI if I don't want to use it.
- Use AI to help create a budget in a style based on my financial situation

Community/Groups

- Create a group with friends.
- Join a community/group with other people, my educator, or my personal finance partner.

User Stories

As a *Young Adult*, I would like to...

Gamification

- Earn achievements for handling my finances.
- Stay on track of my finances through goals.

User Stories

As a **Tester**, I would like to...

- As a tester, I would like to verify that the personal financial dashboard correctly visualizes user spending and debt progress so that the data is accurately represented.
- As a tester, I would like to test the Ai-powered recommendations to ensure that they are relevant and helpful for different financial situations.
- As a tester, I would like to validate that the “what-if” analysis accurately calculates projected financial outcomes based on different inputs.
- As a tester, I would like to simulate various user behaviors to ensure that the app provides appropriate alerts and reminders.

Real World Product vs Prototype

*Fully implemented

*Partially implemented

*Eliminated

Functionality Group	Feature	RWP	Prototype	Notes
Account Management	Add/Delete Account	Full add/delete functionality for user accounts	Fully Implemented	Can use stand-in accounts
	Login/Authentication	MFA and session management	Fully Implemented	Use stand-in accounts
	Profile Management	Customizable user profile	Fully Implemented	Use stand-in accounts
Debt Tracking	Add/Edit Debt	Comprehensive tracking editing	Fully Implemented	
	Smart Payment Reminders	Customizable with escalation	Eliminated	Unsure of method used to send reminders to user without Google APIs
AI-assisted Budgeting	Personalized Budget Creation	AI-driven recommendations	Fully Implemented	Focus generative AI use in what-if analysis
	Expense Tracking	Detailed tracking with visuals	Fully Implemented	
	Budget Adjustments	AI-driven recommendations based on spending patterns	Eliminated	AI recommendations for budgeting too costly/difficult to implement using tokens and OpenAI

Real World Product vs Prototype Cont.

*Fully implemented
*Partially implemented
*Eliminated

Functionality Group	Feature	RWP	Prototype	Notes
Goal Tracking	Short/Long-Term Goals	Customizable short-term goal tracking	Fully Implemented	
	Goal Progress Visualization	Visual tracking	Fully Implemented	
What-If Analysis	AI-assisted projection/simulation	Use of AI for what-if	Fully Implemented	
	Create/Simulate scenarios	Algorithmic Impact simulations	Fully Implemented	
Analytics	Comprehensive Financial Dashboard	Budget and debt visualizations, goal progression,	Fully Implemented	
Reward System	Badges and Rewards	Custom badges for achievements	Partially Implemented	Basic badges for demonstration
	Community/Group page	Full community functionality with leaderboards	Partially Implemented	Can create groups
Financial Literacy Tools	Educational Resources	Upload external resources, show financial concepts embedded in other features	Partially Implemented	Only shows embedded finance information.

User Stories to Tasks

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Debt Tracking	Create/Edit debt source	Create and edit a debt source with source name, amount due, interest rate, term length, and more	Young adults
	Delete debt source	Delete a debt source	Young adults
	Create/edit debt transaction	Create and edit a new debt transaction	Young adults
	Delete debt transactions	Delete a debt transaction	Young adults
	View debt sources and transactions	See all my debt information condense into one easily accessible place	Young adults
	Create pop up screen asking for confirmation	Confirm payment of a debt once I paid the actual source in real life.	Young adults
AI-assisted Budgeting	Integrate gpt-4o-mini to budget creation	Use AI to help create a budget in a style based on my financial situation	Young adults
	Include option to use standard budget during creation process	Create a budget without the use of AI if I don't want to use it	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
AI-assisted Budgeting	CRUD a transaction	Create, edit, and delete an expense/income transaction	Young adults
	CRUD a category	Create, edit and delete an expense/income category	Young adults
	View expenses/income	See the total amount of money I've spent on a category	Young adults
	AI-driven budget recommendations	Receive AI-driven budgeting suggestions, so that I can optimize my spending and savings based on my financial situation	Young adults
	View the budget	View the budget that I created with or without the use of OpenAI model	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Goal Tracking	View goals	Track my short-term and long-term goals in one place	Young adults
	Update goals automatically	Automatically update my goals based on transactions logged into the app	Young adults
	Manual confirmation of goal completion	Press a button confirming I have completed a goal	Young adults
	Record and view past goals	View all my completed goals	Young adults
	CRUD Short-Term/ Long-term Goals	Have short-term goals so I can keep track of my personal finances in small milestones	Young adults
		Have long-term goals so I can set big financial milestones	Young adults
What-If Analysis	Integrate gpt-4o-mini chat box	Use OpenAI model to ask for specific scenarios that the app won't be able to answer	Young adults
	Have gpt-4o-mini read results of what-if analysis, generate response	Use OpenAI model to provide recommendations based on results of the algorithmic scenario.	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
What-If Analysis	Show graphs of results	See graphs that shows the amount of money I save in a period of time	Young adults
		See a graph that shows future debt payments over a period of time.	Young adults
	Implement algorithmic what-if analysis with calculations	Be able to perform basic predictions on savings and debt without the use of OpenAI model	Young adults
Financial Dashboard	View updated financial information	See my debt to income ratio to help me not overextend myself	Young adults
		See my current total spending for this month	Young adults
	Generate trends based on user data with gpt-4o-mini	See current trends, pattern, and statistics generated by OpenAI model from analyzing my data.	Young adults
	View goals	See a short-term goal, long-term goal, and progress for each of them	Young adults
	View achievements	See my last completed achievement	Young adults
		See an achievement that is the closest to completion	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Financial Dashboard	Access other pages from Dashboard	Go to the My Finances Page from the dashboard	Young adults
		Go to the Rewards page from the dashboard	Young adults
		Go to my Debt Table from the dashboard	Young adults
		Go to my goals section from the dashboard	Young adults
Gamification	Update achievement progress	Earn achievements by completing certain requirements/milestones	Young adults
	Design badge tiers	Have the badges level up one tier when I earn it	Young adults

User Stories to Tasks cont.

*Done
*WIP
*Planned

Functionality Group	Task	User Story	User Role
Community/Group	Link accounts to a group	Send/Accept Friend Requests	Young adults
		Join a group with my friends	Young adults
	CRUD a group	Create a group	Young adults